BEING AFFINE IS FPQC LOCAL ON THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The author's parenthetical '(verification omitted, compare with Descent, Lemma descent-lemma-descent-data-sheaves)' inside the descent-datum step is kept verbatim: it omits only the routine cocycle check of \varphi, with the schemes-side comparison given, while the main line (effectivity, sheaf condition, isomorphism descent) is fully derived.

Editorial note. The counted span cites lemmas in Descent, Morphisms of Spaces and Properties of Spaces plus same-chapter lemmas, recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Framework lemma for descending properties of morphisms. Stacks Project, Descent on Spaces, Lemma lemma-descending-properties-morphisms. Supplies the criterion structure (assumptions (1), (2)) that the proof verifies.
- Affineness is local in the \'etale topology. Stacks Project, Morphisms of Spaces, Lemma spaces-morphisms-lemma-affine-local. Supplies assumptions (1) and (2).
- Effectivity of affine descent data. Stacks Project, Descent, Lemma descent-lemma-affine (with Definition descent-definition-descent-datum). Supplies the scheme Y \to Z affine from the descent datum.
- Algebraic spaces satisfy the fpqc sheaf condition. Stacks Project, Properties of Spaces, Proposition spaces-properties-proposition-sheaf-fpqc. Supplies the morphism Y \to X from the descent datum element.
- fpqc descent of the isomorphism property. Stacks Project, Descent on Spaces, Lemma lemma-descending-property-isomorphism. Supplies that Y \to X is an isomorphism from its base change being one.

Source reference. Original lemma 041J: sources/stacks-chow/tags.txt, source lines 5294–5294. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 03WG: sources/stacks-chow/tags.txt, source lines 5111–5111. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 023V: sources/stacks-chow/tags.txt, source lines 2756–2756. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02W5: sources/stacks-chow/tags.txt, source lines 3792–3792. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Date: stacks-project:spaces-descent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 0245: sources/stacks-chow/tags.txt, source lines 2766–2766. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02L5: sources/stacks-chow/tags.txt, source lines 3396–3396. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0APL: sources/stacks-chow/tags.txt, source lines 13479–13479. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 041Y: sources/stacks-chow/tags.txt, source lines 5309–5309. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *The property $\mathcal{P}(f) =$"f is affine" is fpqc local on the base.*

Proof. We will use Lemma 041J to prove this. Assumptions (1) and (2) of that lemma follow from Morphisms of Spaces, Lemma 03WG. Let $Z' \to Z$ be a surjective flat morphism of affine schemes over S. Let $f : X \to Z$ be a morphism of algebraic spaces, and assume that the base change $f' : Z' \times_Z X \to Z'$ is affine. Let X' be a scheme representing $Z' \times_Z X$. We obtain a canonical isomorphism

$$\varphi : X' \times_Z Z' \longrightarrow Z' \times_Z X'$$

since both schemes represent the algebraic space $Z' \times_Z Z' \times_Z X$. This is a descent datum for $X'/Z'/Z$, see Descent, Definition 023V (verification omitted, compare with Descent, Lemma 02W5). Since $X' \to Z'$ is affine this descent datum is effective, see Descent, Lemma 0245. Thus there exists a scheme $Y \to Z$ over Z and an isomorphism $\psi : Z' \times_Z Y \to X'$ compatible with descent data. Of course $Y \to Z$ is affine (by construction or by Descent, Lemma 02L5). Note that $\mathcal{Y} = \{Z' \times_Z Y \to Y\}$ is a fpqc covering, and interpreting ψ as an element of $X(Z' \times_Z Y)$ we see that $\psi \in \check{H}^0(\mathcal{Y}, X)$. By the sheaf condition for X with respect to this covering (see Properties of Spaces, Proposition 0APL) we obtain a morphism $Y \to X$. By construction the base change of this to Z' is an isomorphism, hence an isomorphism by Lemma 041Y. This proves that X is representable by an affine scheme and we win. □